
CSCE 633: Machine Learning

Boosting

Last Time

- Decision Trees
- Random Forest

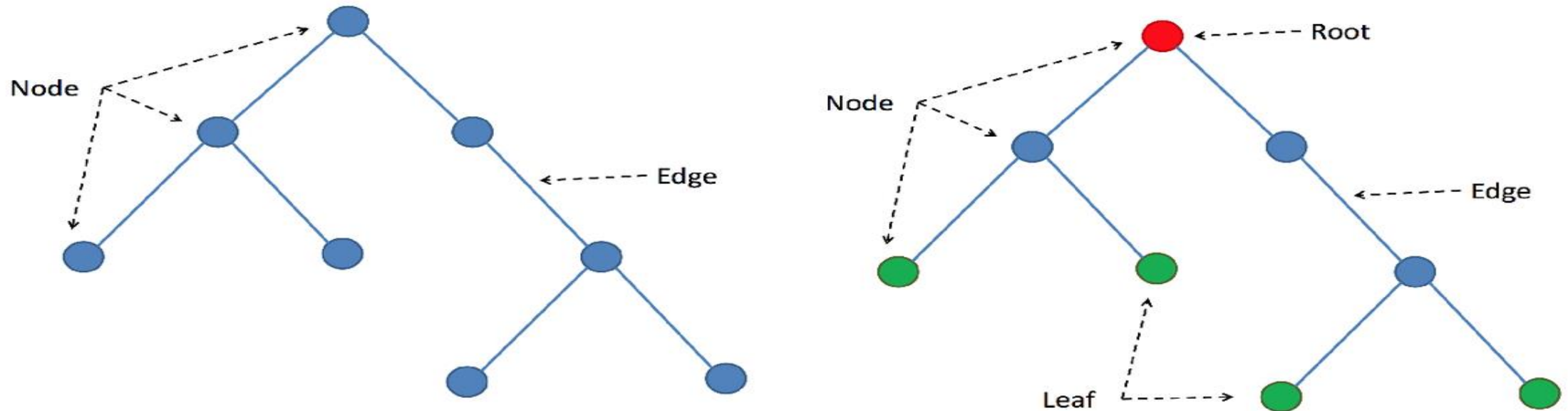
Goals of this lecture

- Review of Decision Trees
- Boosting as an algorithmic concept
- Boosted Decision Trees

Decision Trees

What is a decision tree

A hierarchical data structure implementing the divide-and-conquer strategy for decision making



Can be used for both classification & regression

Gini Index and Entropy

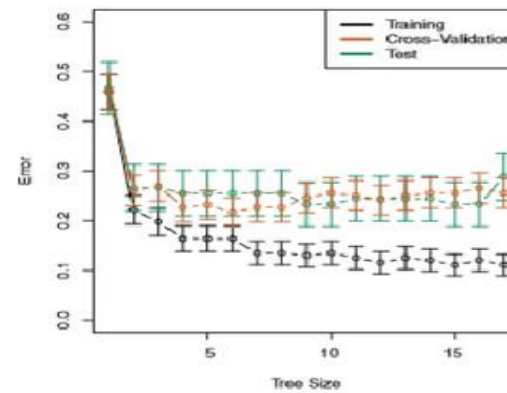
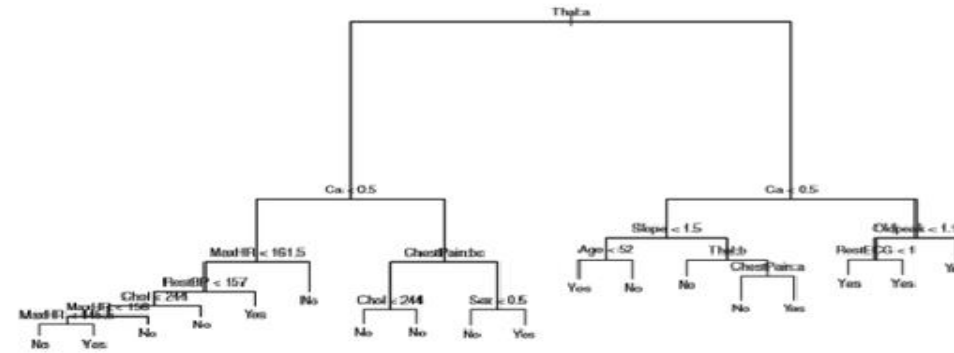
$$G = \sum_{k=1}^K \hat{p}_{mk}(1 - \hat{p}_{mk})$$

, which measures the total variance across K classes. This is a measure of node purity.

$$H = - \sum_{k=1}^K \hat{p}_{mk} \log(\hat{p}_{mk})$$

, Entropy which takes a value near 0 if all the $\hat{p}$ are near zero or one - smaller value if node is pure

Classification and Pruning



Bagging

- Take B different bootstraps of our one dataset
- $\hat{f}_{bag}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}^b(x)$
- Turns out, you can grow these trees without pruning
- Regression - average the values from each tree
- Classification - majority vote from each tree
- Test error can be plotted as a function of B
- B is not a critical parameter (will see shortly) so large B does not mean we overfit

Out of Bag Error

- If we repeatedly fit bootstrapped subsets (say $2/3$ of data)
- Each time we are left with $1/3$ of the data we can call out of bag
- We can estimate error for this - called Out of Bag Estimation

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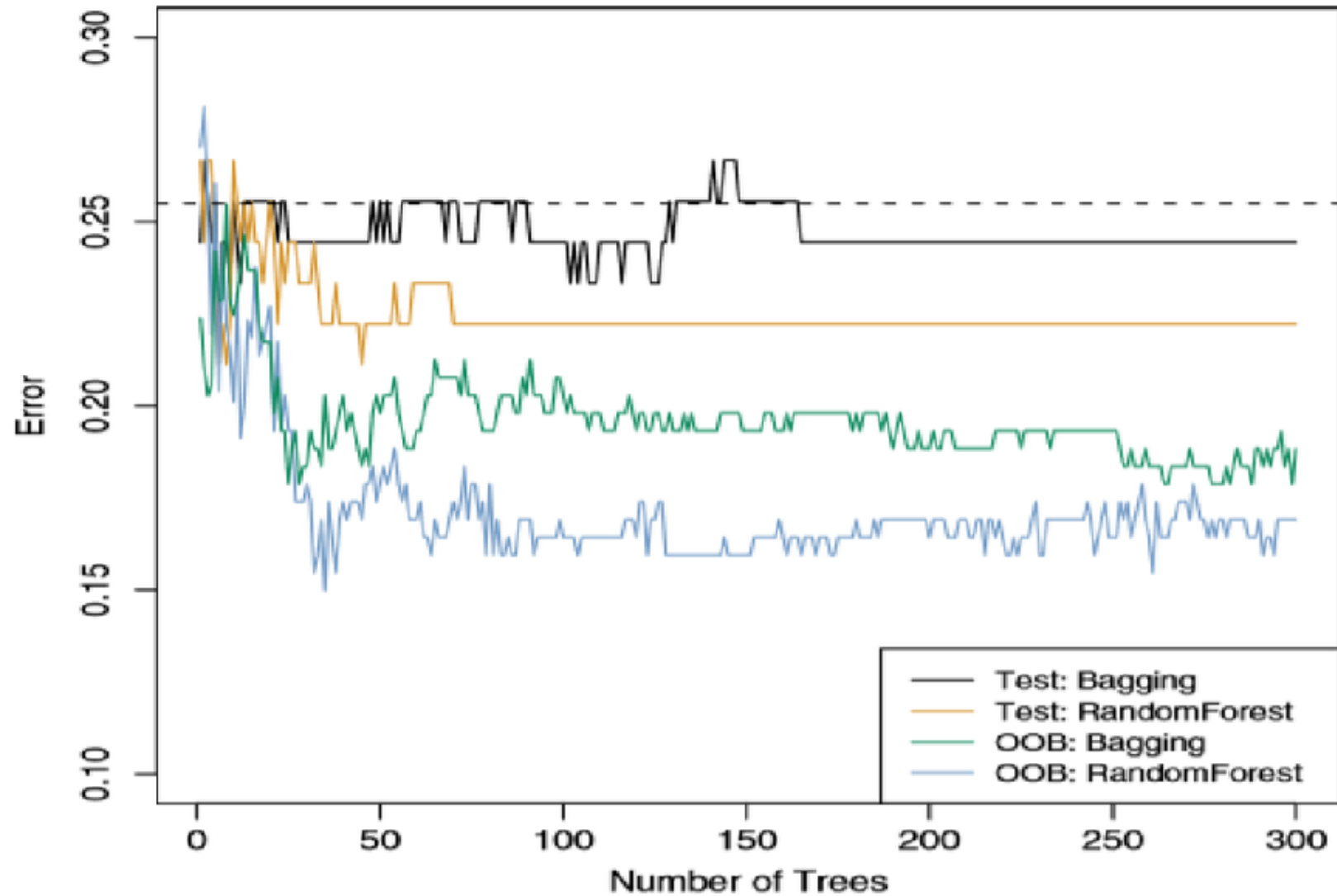
Random Forest

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- other predictors have a chance
- Turns out, this process decorrelates trees

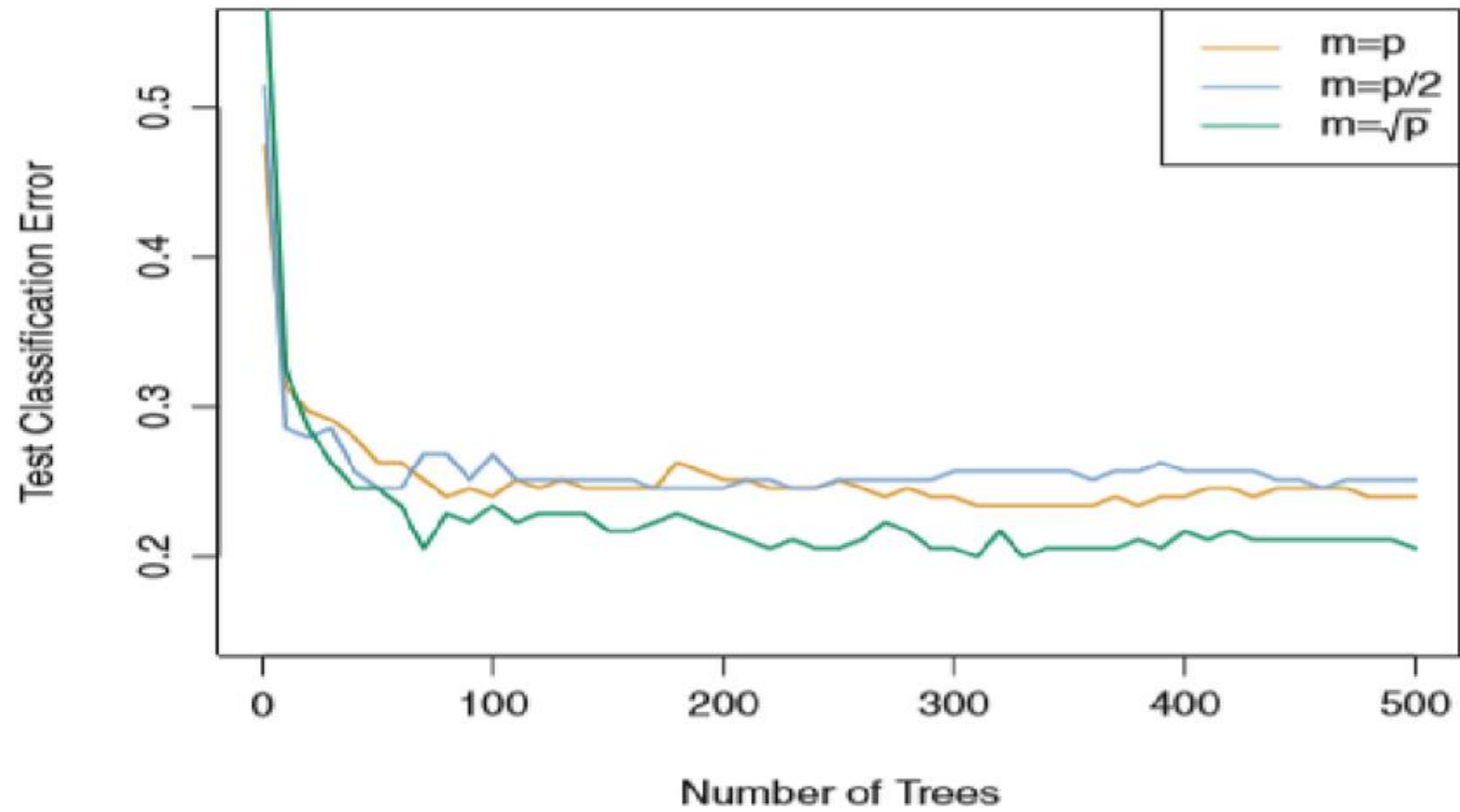
Random Forest

- set $m \approx \sqrt{p}$
- Each time $\frac{p-m}{p}$ predictors aren't even considered
- other predictors have a chance
- Turns out, this process decorrelates trees
- The average tree becomes less variable and thus more reliable

Example: Heart Dataset



RF with different m



Boosting

- Can improve prediction of decision trees even further
- develop a method that can work on any classifier - applied here to regression trees
- Bagging - build each tree randomly
- Random Forest - build each tree randomly, with random variations in predictors allowed
- What if we built trees sequentially?

Takeaways

- Decision Trees and Random Forest work well in generalizing models with lots of trees
- Do not overfit with lots of trees